



Aastik Rajan · Johns Hopkins University · aastikc15@gmail.com · live demo + code: aastikrajan.github.io/pulseguard

What it is. Software that reads the live vital signs during surgery and warns about **4 minutes before** the patient gets into trouble — and says *which* problem (out of 9) and *why*. A research prototype, not a medical device.

Why it matters. Today's monitors only alarm *after* a number crosses a line, and they alarm so often that staff stop trusting them. The one product that predicts ahead (Edwards HPI, in 10,000+ hospitals) predicts only *one* problem, needs an arterial line, and after 8 years still has no trial proving it helps patients.

WHAT I BUILT — ALONE, ON PUBLIC DATA (VITALDB, 5,151 SURGERIES)

Warn before any deterioration	AUPRC 0.454 · AUROC 0.849 · ~4.3 min early
Get the alarms right	62-71% real · vs ~12% in the field
Say <i>which</i> of 9 events is coming	AUROC 0.74-0.96
Work at another hospital, no retraining	AUROC 0.78-0.80 (Seoul→UC-Irvine)
Does low blood pressure <i>cause</i> kidney injury?	mostly no · RR 1.77 → 1.07

What's different. No existing system does all of this together — many events, works across hospitals, honest cause analysis, and trustworthy alarms. And I held it to a strict honesty bar: I locked the test set, threw out a model that was cheating, and reported every failure — including that **hypotension is basically unpredictable** (shown 7 independent ways). The honesty is the point.

One practical upshot. The two signals that matter most are **heart rate and SpO₂ (pulse-ox)** — both cheap. So a low-cost version (ECG + pulse-ox only) could work where arterial lines aren't available.

What I'm looking for. Feedback, a research role, or a collaborator to take the strongest result (works-across-hospitals + the honest benchmark) toward a paper and outside validation.

Retrospective research prototype — not a medical device, diagnosis, or treatment recommendation. Full reproducible benchmark and code at the link above.